

Qatar-2b

R-0973

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_250615 · created 2026-07-29T18:04:14+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460841.6988 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.069 +/- 0.028
Transit depth (Rp/Rs)^2	0.47 +/- 0.38 [%]
Orbital Inclination (inc)	86.75 +/- 1.74
Airmass coefficient 1 (a1)	1.41 +/- 0.01
Airmass coefficient 2 (a2)	-0.215 +/- 0.013
Scatter in the residuals of the lightcurve fit is	0.7 %
Best Comparison Star	None
Optimal Aperture	12.73
Optimal Annulus	15.91
Transit Duration (day)	0.0644 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.069 $\pm$ 0.028 vs 0.16327 (deviation 3.37 σ)
Duration fit vs published	0.0644 d vs 0.0767 d (deviation 2.27 σ)
Mid-time O-C	4.3 min
Depth SNR	2.5
Residual scatter	0.7 %

Light curve

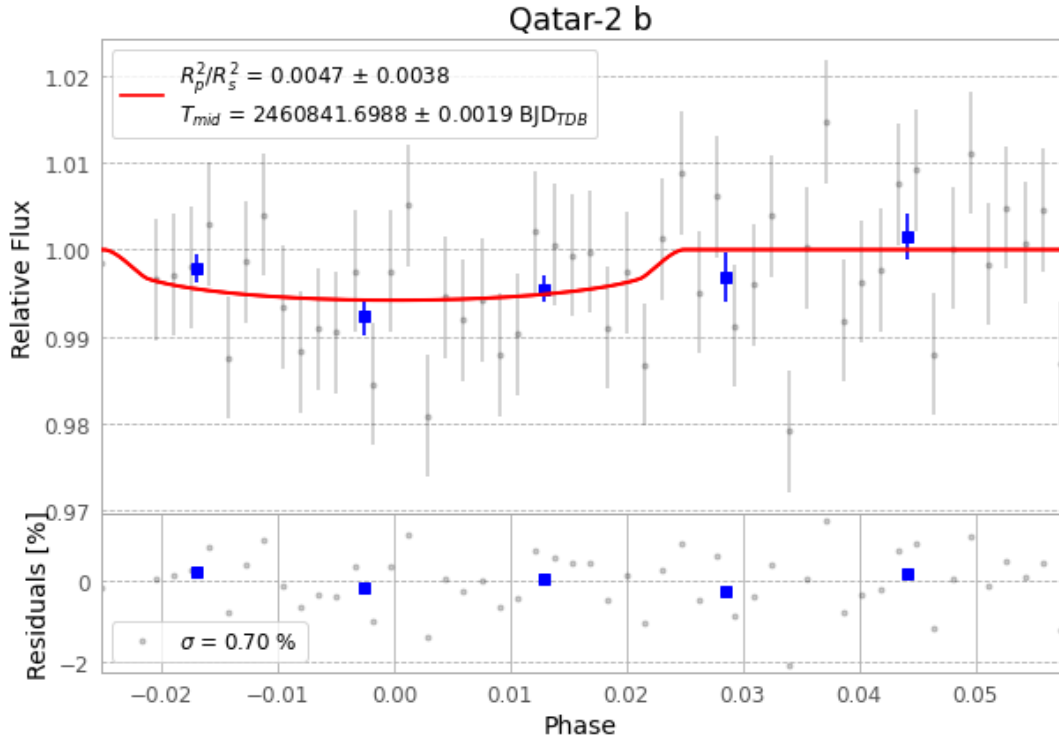


Figure 1 — FinalLightCurve_Qatar-2 b_2025-06-14.png (SHA-256 50897bb5860e1f42...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [457, 416], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 733a8c3a48163a82...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

52 FITS frames (34 MB), Qatar-2250615035114.FITS → Qatar-2250615063015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-250615.log (10f4677c6bca...), FinalLightCurve_Qatar-2 b_2025-06-14.png (50897bb5860e...), FinalLightCurve_Qatar-2 b_2025-06-14.csv (c7965b20ebd3...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 18:04Z.